

RM10-O-M01-BODY

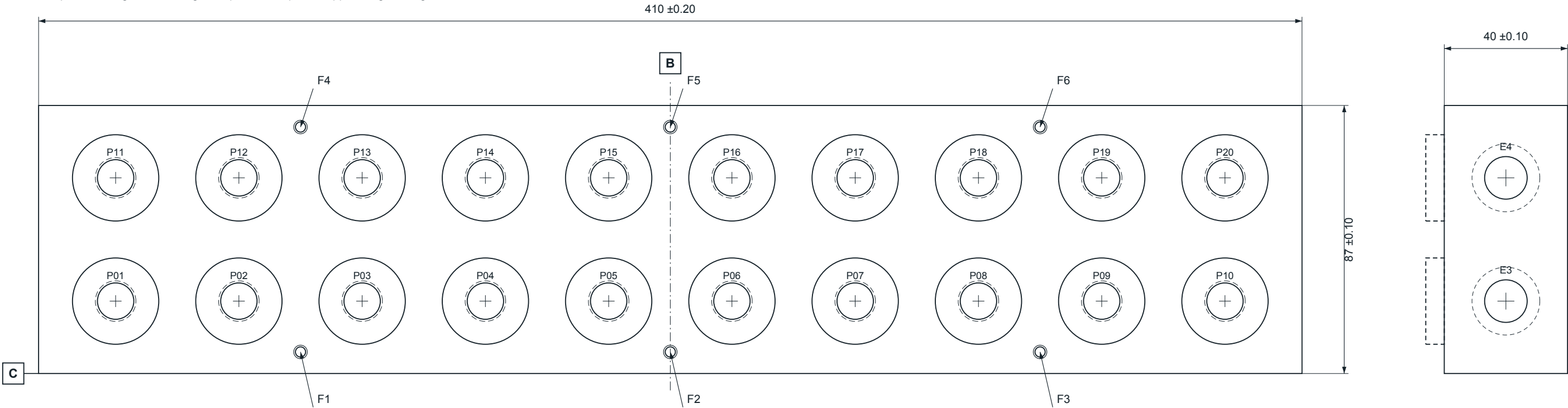
POM MANIFOLD — FRONT AND END PORT / FIXING LOCATIONS

LAYOUT O • DETAIL ISSUE O-M01

Single-part machining drawing

30 TAPPED FEATURES TOTAL: 20 front G 1/4 + 4 end G 1/4 + 6 M4 × 0.7. No thread inserts.

Two uninterrupted Ø11.8 galleries. Long-bore process requires supplier engineering confirmation before manufacture.



FRONT VIEW — looking +Y; countersinks / bosses face the viewer

RIGHT END
looking -X

FRONT PORT IDs	X ±0.10	Z ±0.10
P01 / P11	-180	23.5 / 63.5
P02 / P12	-140	23.5 / 63.5
P03 / P13	-100	23.5 / 63.5
P04 / P14	-60	23.5 / 63.5
P05 / P15	-20	23.5 / 63.5
P06 / P16	20	23.5 / 63.5
P07 / P17	60	23.5 / 63.5
P08 / P18	100	23.5 / 63.5
P09 / P19	140	23.5 / 63.5
P10 / P20	180	23.5 / 63.5

M4 IDs	X ±0.10	Z ±0.10
F1	-120	7
F2	0	7
F3	120	7
F4	-120	80
F5	0	80
F6	120	80

A = body front mounting plane Y0.
B = width midplane X0; C = bottom Z0.
Boss faces: Y-6.0 ±0.10 from A.
Overall depth: 46 REF including bosses.

END ID	FACE	Y / Z ±0.10
E1	LEFT	20 / 23.5
E2	LEFT	20 / 63.5
E3	RIGHT	20 / 23.5
E4	RIGHT	20 / 63.5

Left face: X-205; right face: X+205.
Both ends have the same Y/Z coordinates.
Ø22 dashed rings show sealing lands;
they are NOT recesses or grooves.

20 bosses: Ø28 ±0.10, root R1.0 ±0.10, outer lip C0.5 ±0.10 × 45° ±1° (see sheet 2).
Front drill/thread axes normal to A; end port axes parallel to X. All threads right-hand.
Surface requirements, bore controls, inspection and manufacturing notes: sheet 3. No product markings.

BLACK UNFILLED POM / C OR H

Units: mm • DIN ISO 2768-1 m unless stated

Dimensions at 20 °C • Do not scale drawing

Geometry: approved Revision O

Detail issue: O-M01 • 2026-09-14

Unmodelled helices: machine threads to this drawing

Scale: 0.73:1

Third-angle; views labelled

No product markings

SHEET 1 / 3

PDF + matching STEP

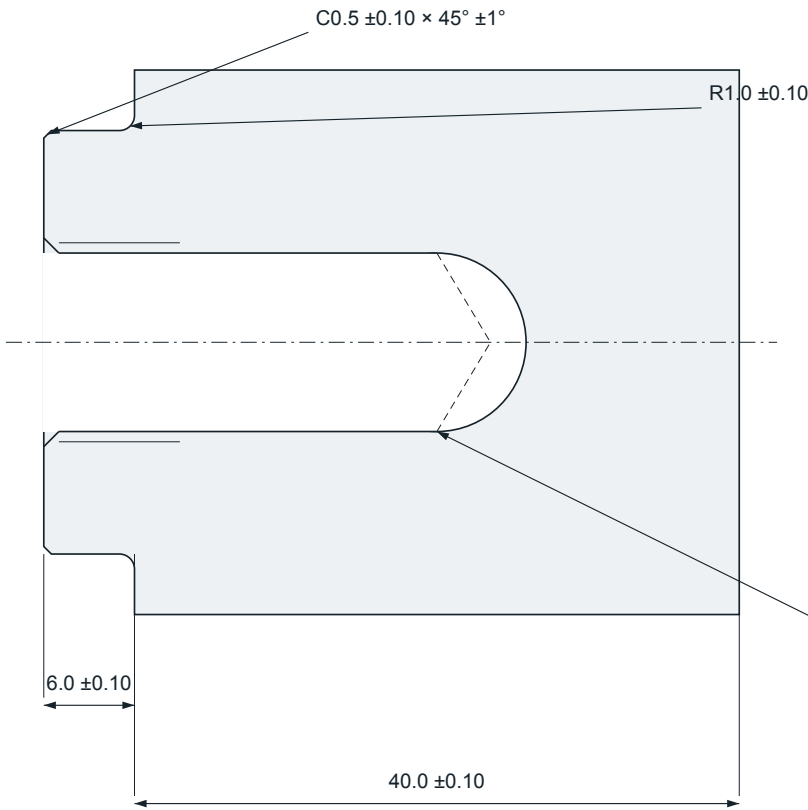
Quotation / supplier DFM

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PORT, BOSS AND BLIND M4 DETAILS

Single-part machining drawing

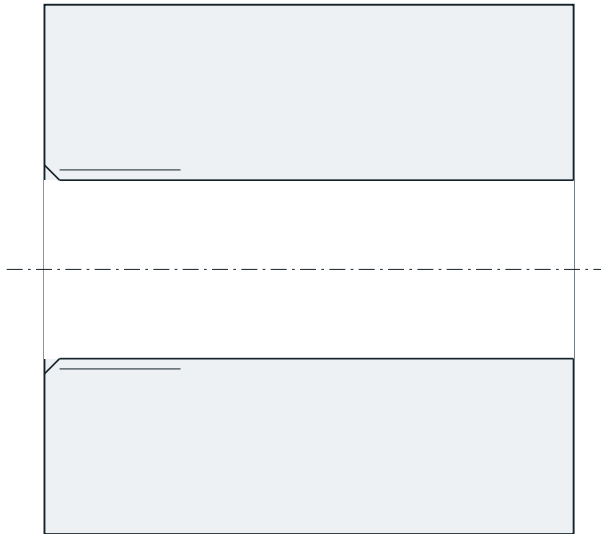
DETAIL A — FRONT G 1/4 PORT / BOSS, 2:1 (20 places)



G 1/4 — DIN EN ISO 228-1, parallel BSPP.
19 TPI (P = 1.33684 REF); 55° thread form.
Entry $\varnothing 13.8 \pm 0.10 \times 90^\circ \pm 1^\circ$ included.
Entry depth 1.0 REF to $\varnothing 11.8$ pilot.
8 MIN full-form thread length after lead-in.
Thread + tool run-out must end within
16 mm of the sealing face.
Front pilot: $\varnothing 11.8$, 26.0 ± 0.20 full diameter
from boss face to gallery axis; 118° tip REF.
Protect flat seal annulus: $\varnothing 13.8$ to $\varnothing 27$ REF.

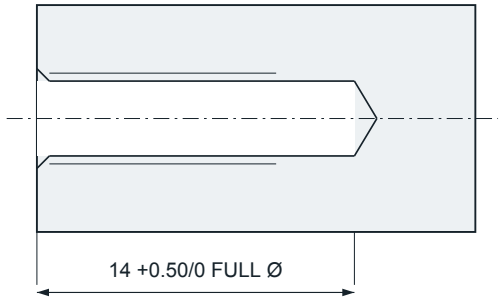
$\varnothing 11.8$ gallery, axis Y20

DETAIL B — END G 1/4, 2:1 (4 places)



Same G 1/4 thread and $\varnothing 13.8 \times 90^\circ$ entry.
8 MIN full-form thread after lead-in.
Thread/run-out ≤ 16 from each end face.
Flat $\varnothing 22$ MIN sealing land on end face.
Unthreaded bore continues to opposite end.
No end caps, inserts, glued plugs or splits.

DETAIL C — M4 × 0.7 - 6H (6 places), 3:1



M4 × 0.7 - 6H, 10 MIN full-form thread after entry.
 $\varnothing 3.3$ tapping pilot, 14 +0.50/0 full-diameter depth.
118° $\pm 2^\circ$ drill point; total tip depth ≤ 15.5 from A.
Entry $\varnothing 4.4 \pm 0.10 \times 90^\circ \pm 1^\circ$ included (0.55 REF).

No inserts. Do not break into a coolant passage.
Drill depth $\neq$ full thread length; leave run-out space.

General unspecified external edges: break 0.10–0.20 max.

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Dimensions at 20 °C • Do not scale drawing

Geometry: approved Revision O

Detail issue: O-M01 • 2026-09-14

Unmodelled helices: machine threads to this drawing

Scale: AS SHOWN

Third-angle; views labelled

No product markings

SHEET 2 / 3

PDF + matching STEP

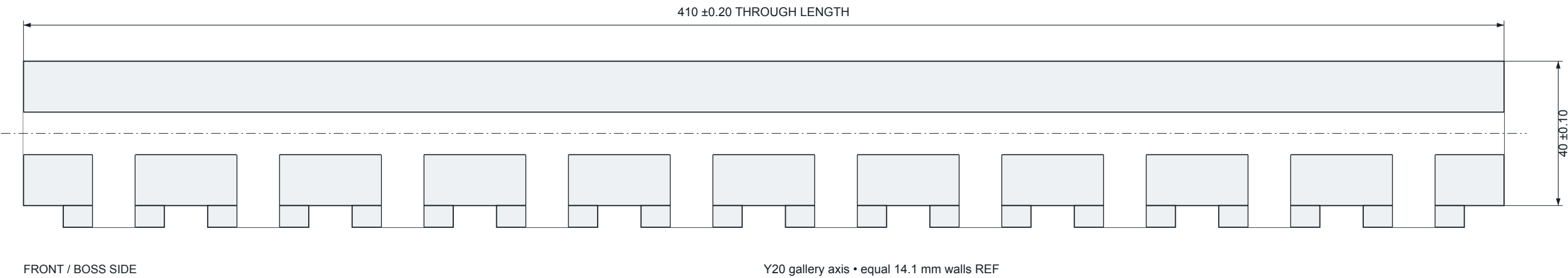
Quotation / supplier DFM

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LONG-BORE SECTION, TOLERANCES AND MACHINING NOTES

Single-part machining drawing

SECTION THROUGH ONE GALLERY AT Z23.5 — second gallery identical at Z63.5



BORES / THREADS / CLEANLINESS

- Two galleries Ø11.8 +0.08/0 through 410 mm, axes Y20; Z23.5 and Z63.5. Continuous parallel networks only.
- Opposed drilling is permissible. Nominal reach 207 mm per end gives 4 mm overlap; select tooling/process.
- Gallery centreline deviation ≤0.30 from nominal axis; meeting step ≤0.30. No blind webs or wall breakthrough.
- Unthreaded gallery / branch bores: Ra ≤6.3 µm.
- G 1/4: gauge to ISO 228-2. M4: gauge to ISO 1502.
Thread quality governs the final tapped pilot diameter.
G is PARALLEL; no NPT, R, Rp or tapered substitute.
- INTERNAL EDGE DEBURRING IS NOT REQUIRED.
Clean / flush out loose machining chips only.
Verify both galleries are continuous and open.
- Deliver dry and clean. No chemical treatment, impregnation or adhesive repairs without agreement. No product marks.

DATUMS / FUNCTIONAL SURFACES

- A: front body mounting plane; flatness 0.15.
B: derived width midplane. C: bottom plane.
Front boss/thread and M4 positions: X/Z ±0.10.
Side mouth positions: Y/Z ±0.10 at each end face.
- Boss OD Ø28 ±0.10; height 6.0 ±0.10; root R1 ±0.10.
Lip C0.5 ±0.10 ×45° ±1°. Do not round the sealing face.
- Each front sealing annulus and each Ø22 end land:
Ra ≤1.6 µm, local flatness 0.05; no radial scratches.
Seal plane perpendicular to its thread axis within 0.05 across Ø22. Do not recess the sealing lands.
- Other machined exterior surfaces: Ra ≤3.2 µm.
- Dimensions apply after material stabilisation at 20 °C.
Declare resin grade and stock form before manufacture.
POM-C or POM-H accepted; supply grade datasheet.
- 2 mm rack-envelope gap is not a tolerance allocation; this drawing does not qualify the assembled rack fit.

SUPPLIER DFM HOLD POINT: confirm long-bore process, alignment inspection and material grade before cutting.

STEP contains nominal tapping pilots, not thread helices. General external edge deburring is not explicitly modelled.
Boss lip chamfers, port-entry cones and M4 pilot drill points are modelled in the matching O-M01 STEP.
Operator approval applies to layout O. This detail issue adds edge / drill specifications; no pressure or structural rating is claimed.

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Dimensions at 20 °C • Do not scale drawing

Geometry: approved Revision O

Detail issue: O-M01 • 2026-09-14

Unmodelled helices: machine threads to this drawing

Scale: SECTION 0.80:1

Third-angle; views labelled

No product markings

SHEET 3 / 3

PDF + matching STEP

Quotation / supplier DFM